Project Proposal: Reinforcement Learning for Human-Like Locomotion Using MuJoCo

Course: Computational Motor Control
Team Members:

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Problem Description:

The mechanism underlying human learning and the generation of the walking behavior is among the key problems in computational motor control. Instead of implementing an inflexible set of movement commands, the brain seems to learn to walk by trial and error, in a manner that resembles how reinforcement learning algorithms work. This project takes inspiration from that biological process and asks: *Can an RL agent, given the right reward signals, independently discover a stable, efficient walking gait inside a physics simulator?*

To answer this, we will use MuJoCo an advanced physics engine that is popular in robotics and biomechanics to simulate a bipedal humanoid model. We will then train our RL agent using PPO algorithm to actuate the joint torque of that body. The reward will be provided to the agent depending upon its forward motion, upright state, and coordinated limb movement, and penalties will be given if it falls or consumes too much energy. With many iterations in simulation, the agent will autonomously learn to walk.

After the training of our walking policy, the gait will be examined and compared against existing rules that govern human locomotion. Questions such as "How does the foot contact the ground?" or "What happens to the body's center of mass over time?" or "How much energy is required in each step?" are among those that will be explored. Should the learning algorithm converge toward the same principles governing human movement – namely the principle of minimal energy cost during locomotion – this will be powerful evidence for such a hypothesis.

This research project lies at the crossroads of biomechanics, neuroscience, and artificial intelligence and addresses the fundamental principles discussed in this course regarding the planning, learning, and computational modeling of movement.

Goals and Deliverables:

By the end of this project, we aim to:

- Set up and configure a MuJoCo humanoid environment (Walker2d-v4 or Humanoid-v4) using the Gymnasium interface.
- Design and tune a biologically motivated reward function that encourages stable, efficient locomotion.
- Train a PPO policy from scratch using the Stable-Baselines3 library.

- Record and visualize the trained agent's gait: joint angles, ground reaction forces, center-of-mass trajectory, and muscle activation proxies.
- Compare the agent's emergent walking strategy to documented human gait patterns and motor control principles.
- Reflect on what the RL results suggest about biological motor learning — where they agree with theory, and where they fall short.

Literature Review:

The following papers form the theoretical and technical backbone of this project. They were selected to cover the RL methods we will apply, the biological context we are drawing from, and direct examples of this kind of work being done in the field.

1. Lee et al. (2021) - *Deep Reinforcement Learning for Modeling Human Locomotion Control in Neuromechanical Simulation*

   This is what forms the fundamental theoretical basis for all our work. In the journal "Journal of NeuroEngineering and Rehabilitation," the authors of the paper demonstrate that Deep Reinforcement Learning is indeed the correct computational framework for modeling human motor control, particularly in terms of modeling high-level behavior such as gait transition, awareness of environment, and planning skills that cannot be modeled using classical reflexive mechanisms or central pattern generators (CPGs). They achieve this by demonstrating the power of reinforcement learning via the "Learn to Move" NeurIPS competition that took place between 2017 and 2019, where more than 1,300 teams trained RL agents on a 3D musculoskeletal model to walk without any reference motions.

2. Schulman et al. (2017) - *Proximal Policy Optimization Algorithms*

   We have decided to implement the PPO algorithm for reinforcement learning. This paper explains the advantages of using PPO as a reliable alternative to previous policy-based reinforcement algorithms. It provides an in-depth insight into how PPO balances implementation simplicity and efficiency in practice while being extensively used for locomotion problems. We will gain detailed knowledge of why PPO is efficient, what parameters are included in PPO, and how to analyze possible convergence issues during training.

3. Wang et al. (2022) - *MyoSim: Fast and Physiologically Realistic MuJoCo Models for Musculoskeletal Motor Control*

   This paper is most like our work. The authors create biologically realistic models of human limbs within MuJoCo using muscle models of the Hill type and train reinforcement learning agents to actuate them. Their findings show that reinforcement learning agents are capable of learning movement policies that mimic human movement when trained in biologically realistic simulations. This paper provides us with both a technical blueprint and a benchmarking tool for our results.

4. Dembia et al. (2020) - *OpenSim Moco: Musculoskeletal Optimal Control*
   In this paper, we introduce Moco, a direct collocation optimization library implemented on top of OpenSim as another way to solve the problem of optimal motion planning. Although in our project, we take a different direction by utilizing RL (which is data-driven) compared to optimal control (which is mathematics-driven), it is insightful for us to learn about Moco as it allows us to see how an optimal way of walking can be derived mathematically, thus giving us a reference point for comparison with the result generated by RL.

Technical Approach:

- Tools and Libraries -

Tools	Purpose
MuJoCo (DeepMind, free)	Physics simulation of the humanoid body
Gymnasium (Walker2d-v4)	Standard RL environment interface
Stable-Baselines3 (PPO)	RL training framework
MyoSuite (optional)	Physiologically realistic muscle models
NumPy / Matplotlib	Data processing and gait analysis
Python 3.10+	Primary programming language

- Reward Function Design -

  The reward function serves as the core of our project. The ill-conceived reward may generate unnatural gait behaviors like hopping and spinning despite the agent succeeding. We intend to develop a reward function with the following elements:

  - Forward velocity reward: Encourages the agent to move in the target direction
  - Upright posture reward: Penalizes excessive trunk lean using torso angle
  - Energy penalty: Penalizes large joint torques to push the agent toward efficient movement.
  - Foot contact reward: Provides a small bonus for heel-to-toe foot strike patterns
  - Fall penalty: A large negative reward if the agent's torso drops below a height threshold

- Training Protocol -
  The training process will last approximately 5-10 million environment timesteps, which should be computationally viable on the CPU of a current laptop within 4-8 hours, using the PPO algorithm provided by the Stable-Baselines3 library. The results will be stored every 500,000 timesteps, and the policy will be visualized at every checkpoint.

References and Resources:

1. Lee et al. (2021), *Deep Reinforcement Learning for Modeling Human Locomotion Control in Neuromechanical*
2. SimulationSchulman et al. (2017), *Proximal Policy Optimization Algorithms*
3. Wang et al. (2022), *MyoSim: Fast and Physiologically Realistic MuJoCo Models*
4. Dembia et al. (2020), *OpenSim Moco: Musculoskeletal Optimal Control*
5. OpenSim RL project -RL applied to OpenSim musculoskeletal models, a related prior effort